20 The density of apple juice is 1.05 grams per cm^3.

The density of fruit syrup is 1.4 grams per cm^3.

The density of carbonated water is 0.99 grams per cm^3.

25 cm^3 of apple juice are mixed with 15 cm^3 of fruit syrup and
280 cm^3 of carbonated water to make a drink with a volume of 320 cm^3.

Work out the density of the drink.
Give your answer correct to 2 decimal places.

Each table scales volume in cm^3 to mass in grams.

Apple juice

	cm3	g	
×25	1	1.05	×25
	25	26.25	

Fruit syrup

	cm3	g	
×15	1	1.4	×15
	15	21	

Carbonated water

	cm3	g	
×280	1	0.99	×280
	280	277.2	

1 mark
Find at least one mass

Total mass = 26.25 + 21 + 277.2 = 324.45 g

1 mark
Add all three masses

1 mark
Round to 2 decimal places

Density = 324.45 ÷ 320 = 1.01390625 g/cm^3

1 mark
Total mass ÷ total volume

1.01 g/cm^3

(Total for Question 20 is 4 marks)